



8086 Architecture

Block diagram of 8086

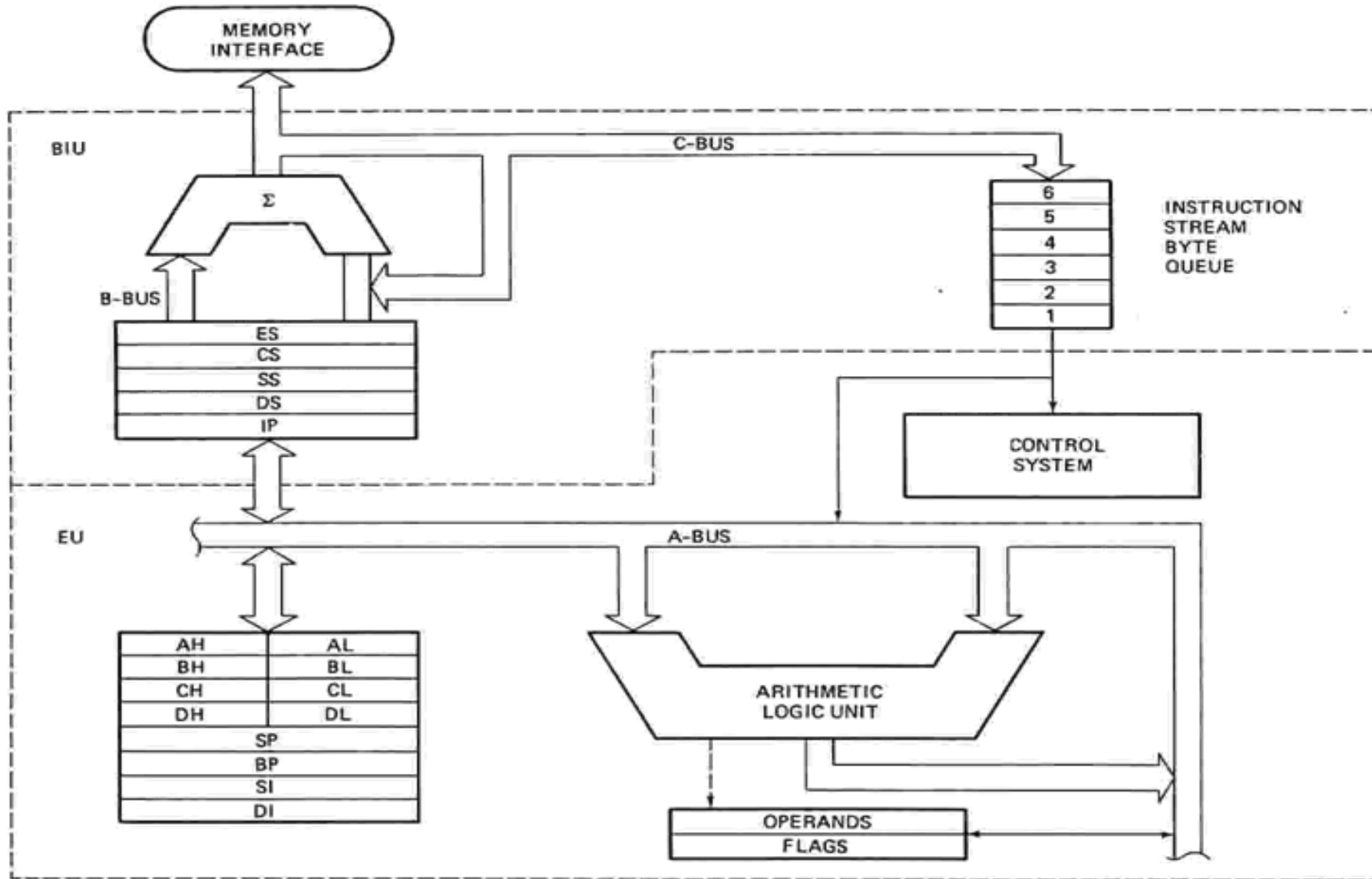
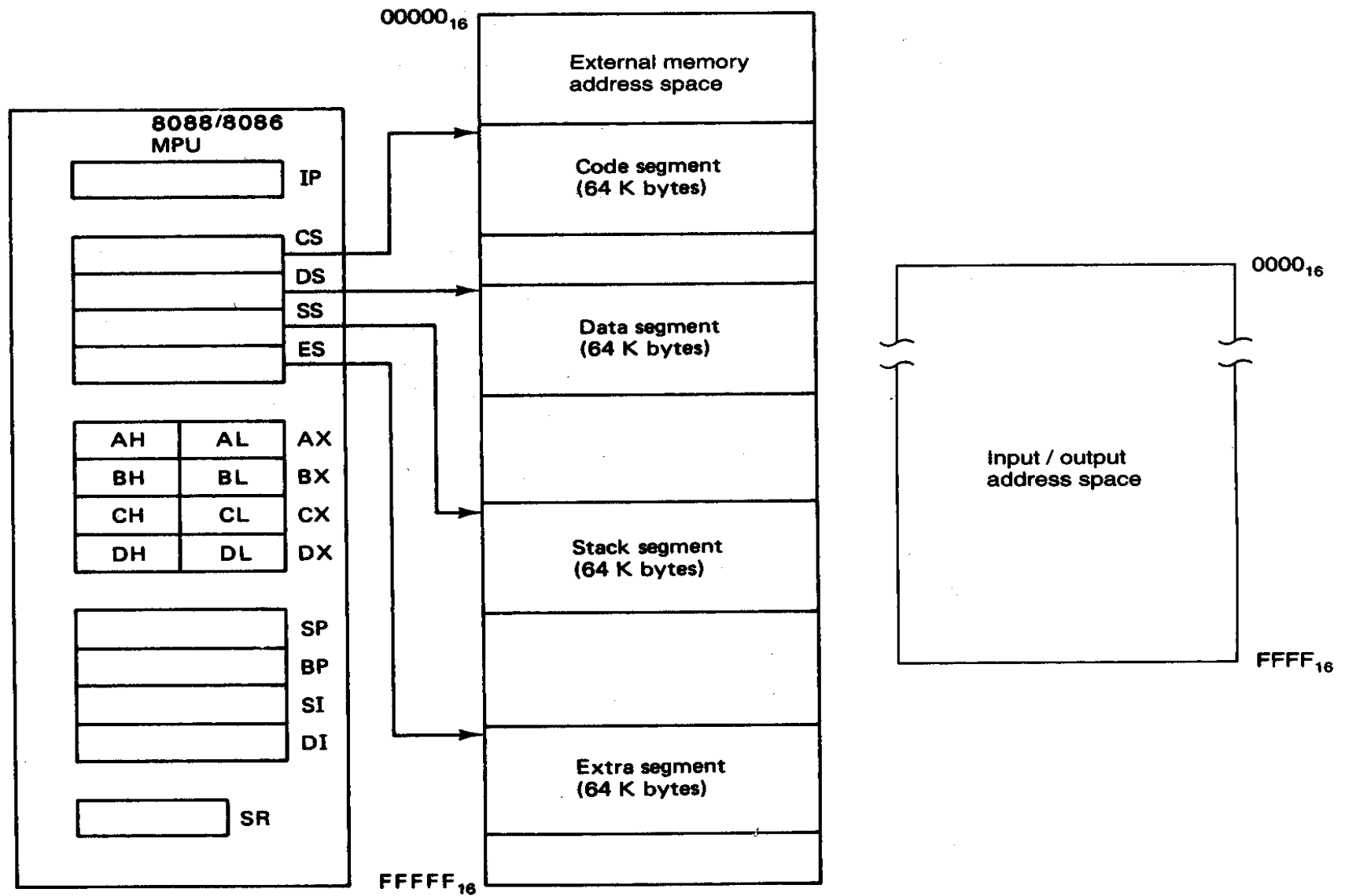


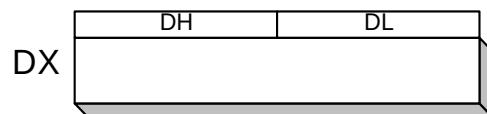
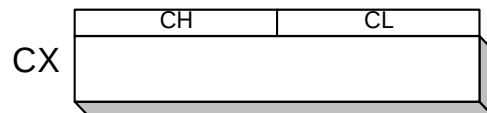
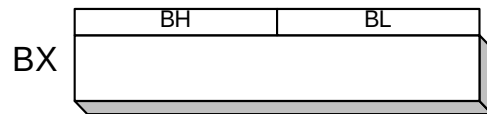
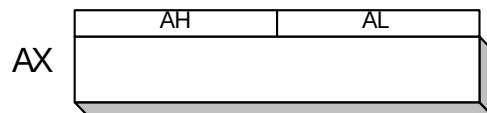
FIGURE 2-7 8086 internal block diagram. (Intel Corp.)

Software Model of the 8086 Microprocessors



8086

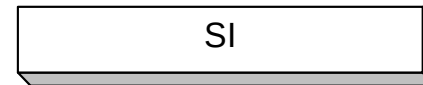
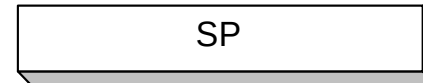
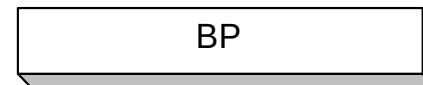
General Purpose



Status and Control



Index



Segment



15	H	8	7	L	0
AX (Accumulator)					
AH			AL		
BX (Base Register)					
BH			BL		
CX (Used as a counter)					
CH			CL		
DX (Used to point to data in I/O operations)					
DH			DL		

gisters

AX - the Accumulator
BX - the Base Register
CX - the Count Register
DX - the Data Register

- Normally used for storing temporary results
- Each of the registers is 16 bits wide (**AX, BX, CX, DX**)
- Can be accessed as either 16 or 8 bits AX, AH, AL

General Purpose Registers

- **AX**

- Accumulator Register
- Preferred register to use in arithmetic, logic and data transfer instructions because it generates the shortest Machine Language Code
- Must be used in multiplication and division operations
- Must also be used in I/O operations

- **BX**

- Base Register
- Also serves as an address register

General Purpose Registers

- **CX**

- Count register
- Used as a loop counter
- Used in shift and rotate operations

- **DX**

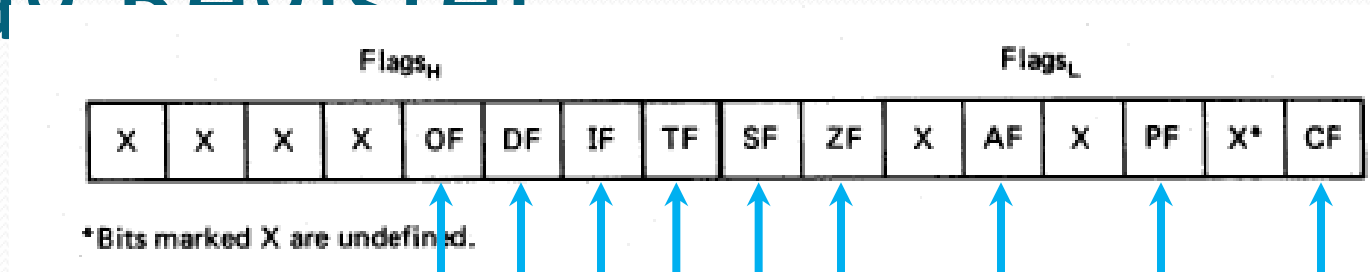
- Data register
- Used in multiplication and division
- Also used in I/O operations

Pointer and Index Registers

SP	Stack Pointer
BP	Base Pointer
SI	Source Index
DI	Destination Index
IP	Instruction Pointer

- All 16 bits wide, L/H bytes are not accessible
- Used as memory pointers
 - Example: MOV AH, [SI]
 - *Move the byte stored in memory location whose address is contained in register SI to register AH*
- IP is not under direct control of the programmer

Flag Register



Overflow

Direction

Interrupt enable

Trap

Sign

Zero

Auxiliary Carry

Parity

Carry

6 are status flags
3 are control flag

8086 Processor Model

BIU registers
(20 bit adder)

ES	Extra Segment
CS	Code Segment
SS	Stack Segment
DS	Data Segment
IP	Instruction Pointer

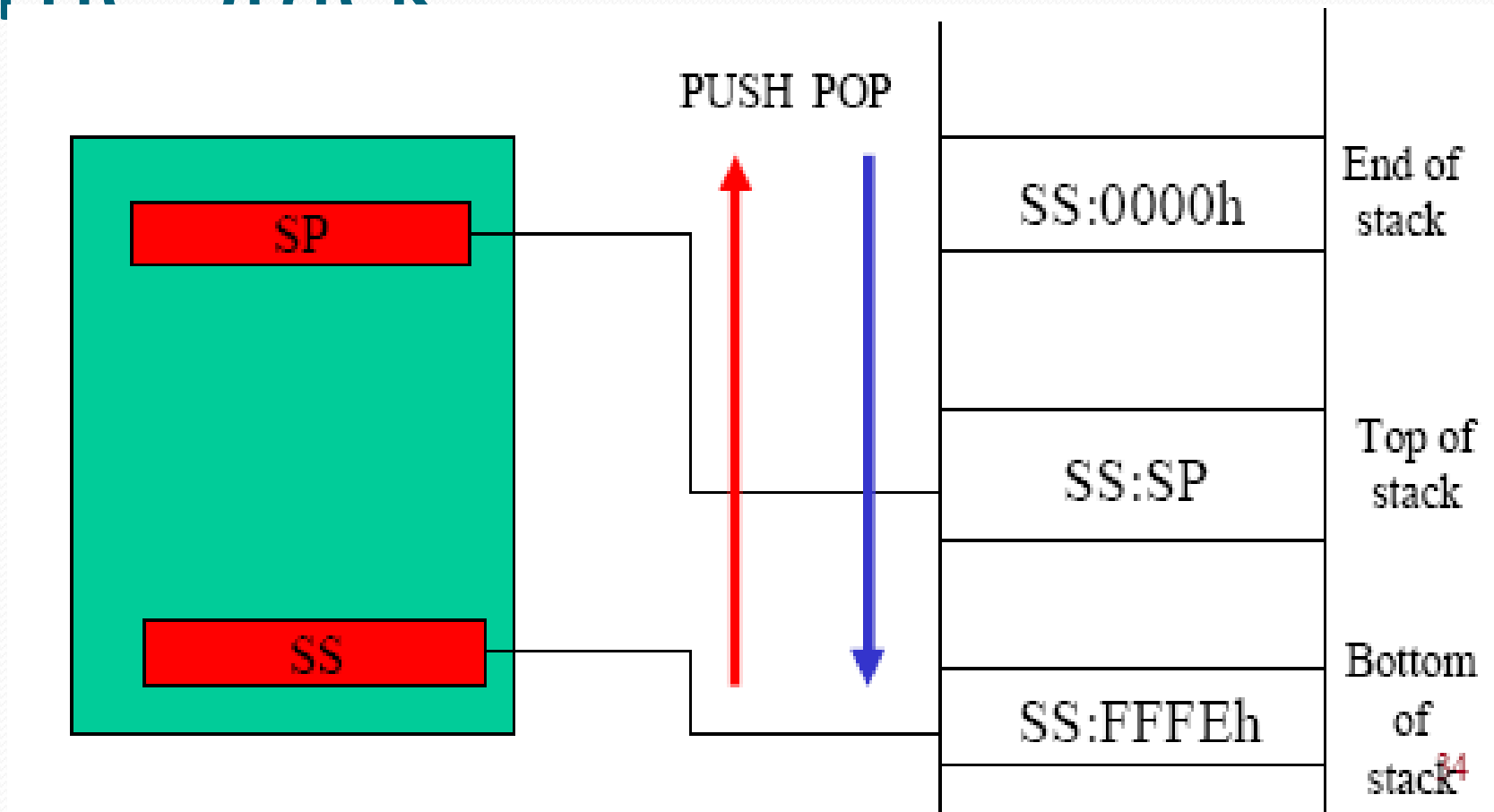
EU registers

AX	AH	AL	Accumulator
BX	BH	BL	Base Register
CX	CH	CL	Count Register
DX	DH	DL	Data Register
	SP		Stack Pointer
	BP		Base Pointer
	SI		Source Index Register
	DI		Destination Index Register
	FLAGS		

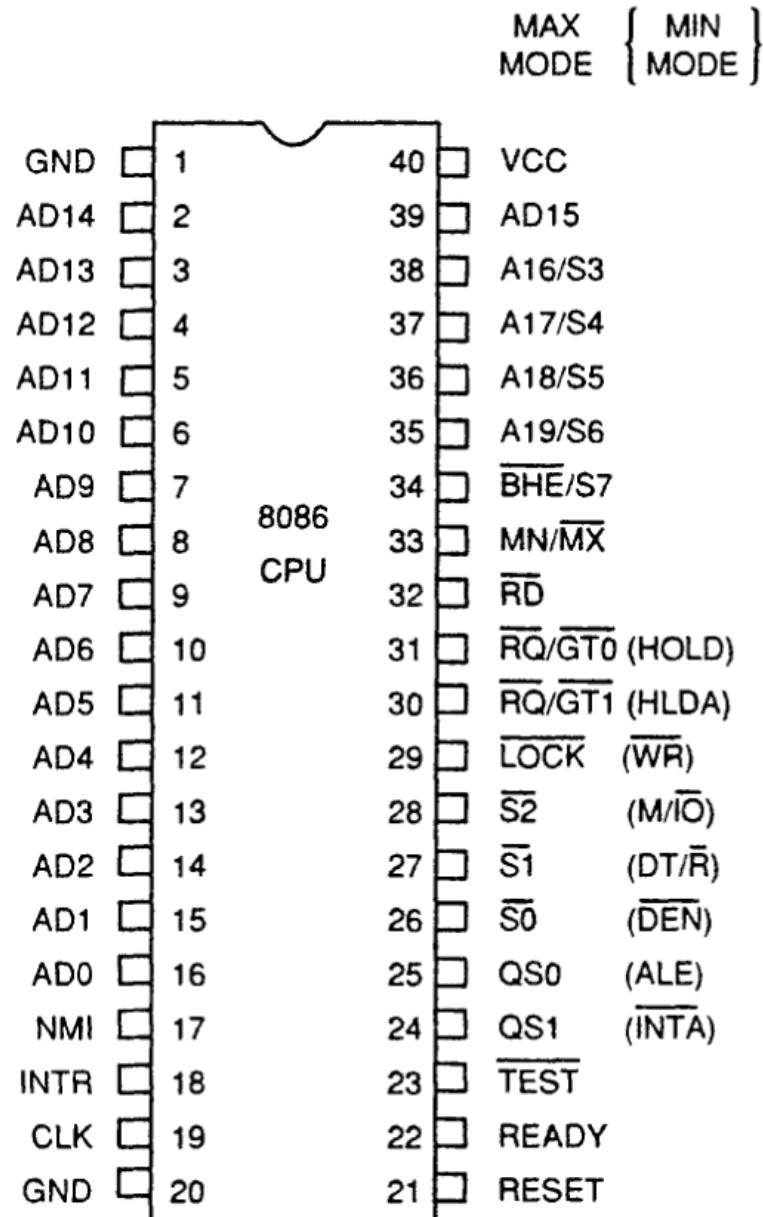
The Stack

- The stack is used for temporary storage of information such as data or addresses.
- When a **CALL** is executed, the 8086 automatically **PUSH**es the current value of CS and IP onto the stack.
- Other registers can also be pushed
- Before return from the **subroutine**, **POP** instructions can be used to pop values back from the stack into the corresponding registers.

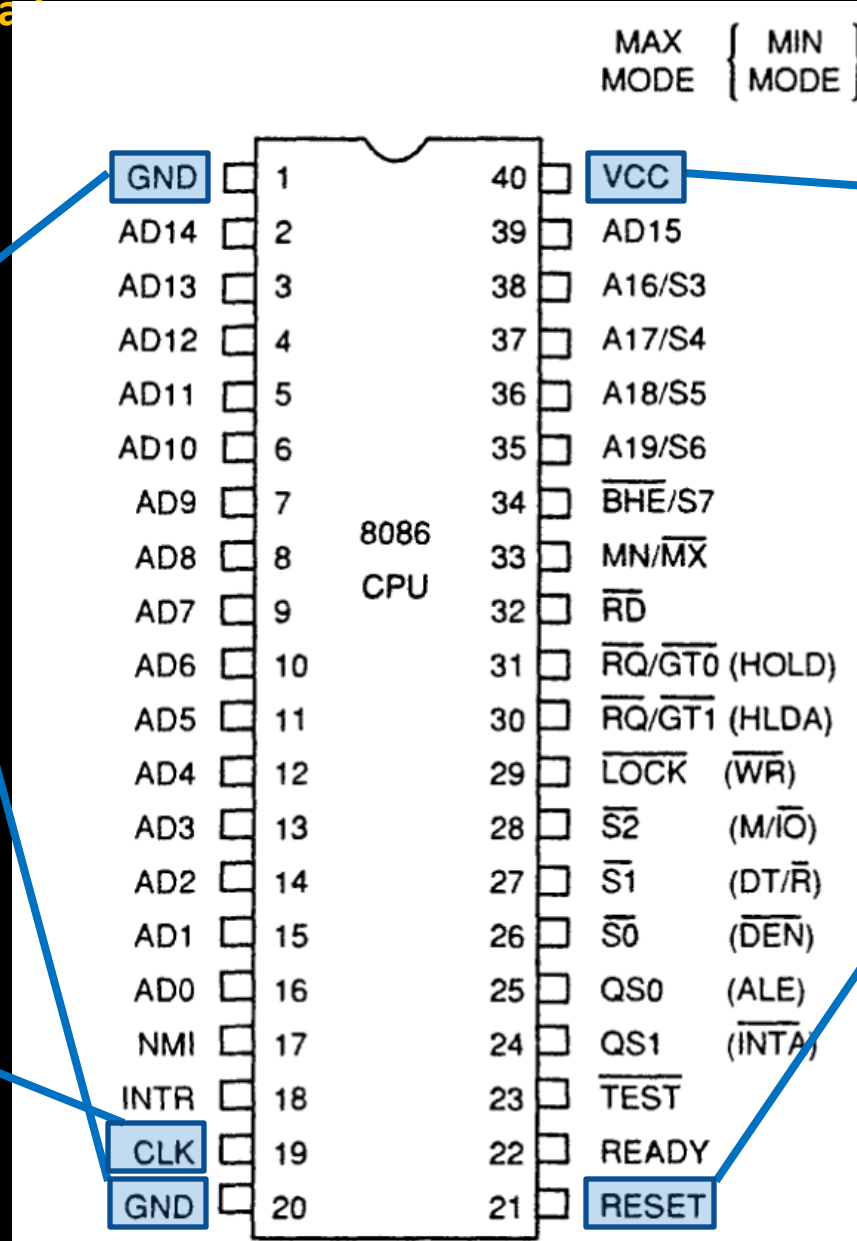
The Stack



INTEL 8086 - Pin Diag



INTEL 8086 - Pin Details



Power Supply

5V $\pm$ 10%

Ground

Reset

Registers, seg
regs, flags

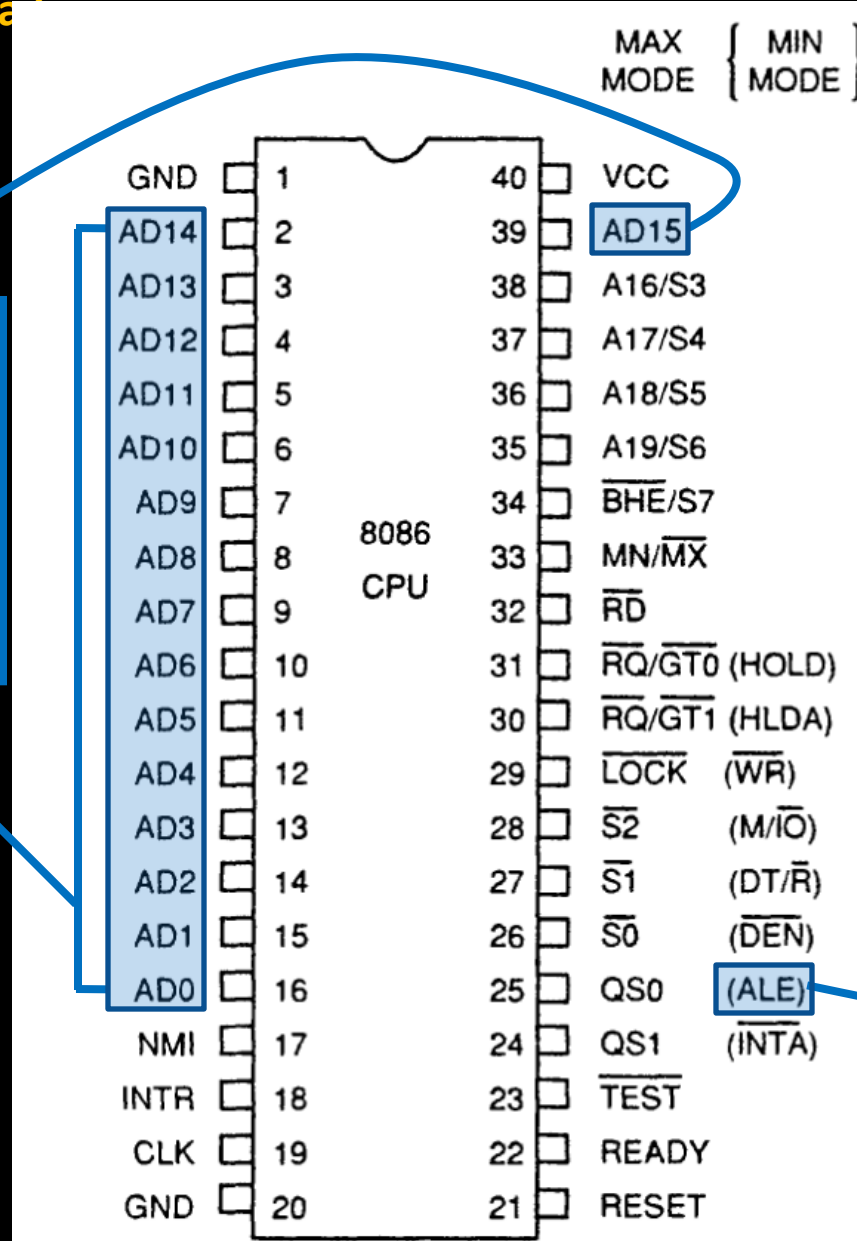
CS: FFFFH, IP:
0000H

If high for
minimum 4
clks

Clock

Duty cycle: 33%

INTEL 8086 - Pin Details



Address/Data Bus:

Contains address bits $A_{15}-A_0$ when ALE is 1 & data bits $D_{15}-D_0$ when ALE is 0.

Address Latch Enable:

When high, multiplexed address/data bus contains address information.

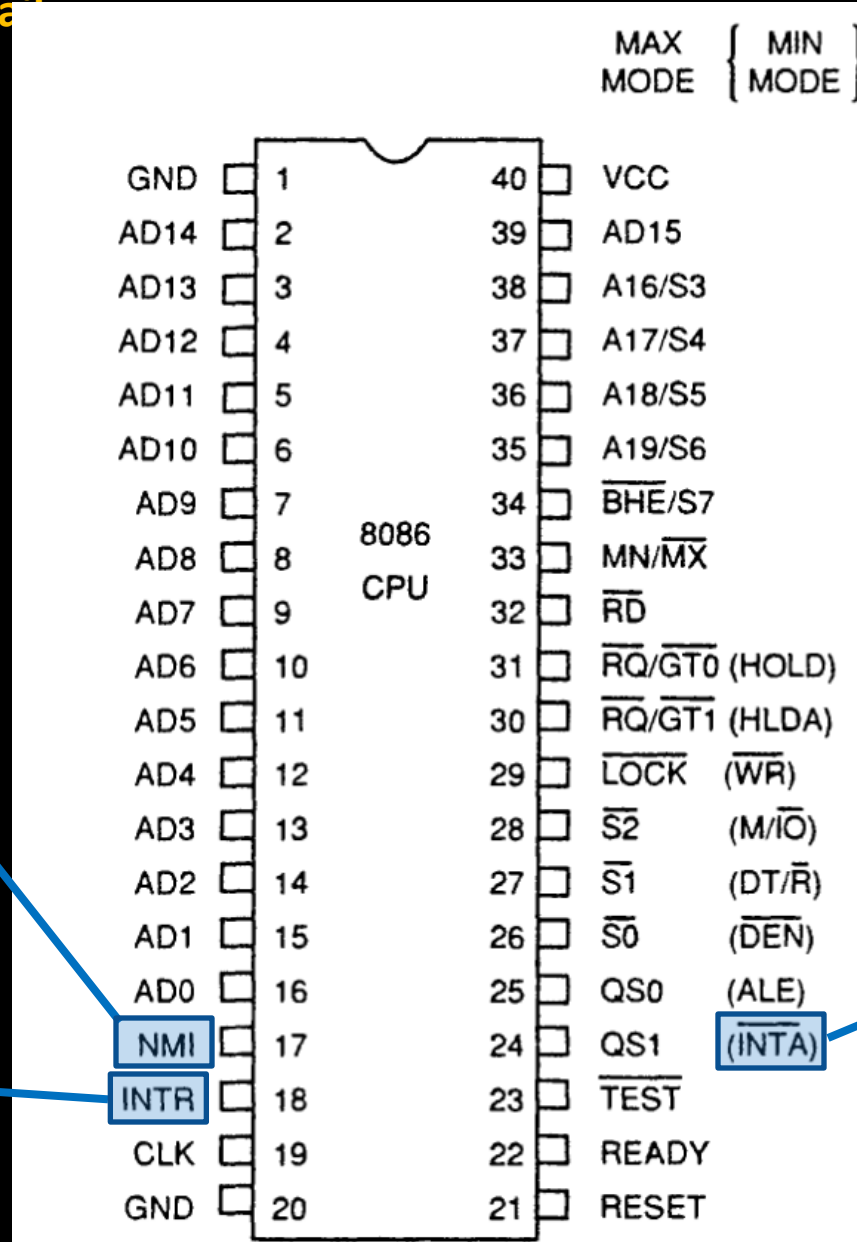
INTEL 8086 - Pin Details

INTERRUPT

Non - maskable
interrupt

Interrupt request

Interrupt
acknowledge



INTEL 8086 - Pin Details

					MAX MODE	{ MIN MODE }
GND	1		40	VCC		
AD14	2		39	AD15		
AD13	3		38	A16/S3		
AD12	4		37	A17/S4		
AD11	5		36	A18/S5		
AD10	6		35	A19/S6		
AD9	7		34	$\overline{\text{BHE}}/\text{S7}$		
AD8	8	8086 CPU	33	$\text{MN}/\overline{\text{MX}}$		
AD7	9		32	$\overline{\text{RD}}$		
AD6	10		31	$\overline{\text{RQ}}/\overline{\text{GT0}}$ (HOLD)		
AD5	11		30	$\overline{\text{RQ}}/\overline{\text{GT1}}$ (HLDA)		
AD4	12		29	$\overline{\text{LOCK}}$ ($\overline{\text{WR}}$)		
AD3	13		28	$\overline{\text{S2}}$ ($\text{M}/\overline{\text{IO}}$)		
AD2	14		27	$\overline{\text{S1}}$ ($\text{DT}/\overline{\text{R}}$)		
AD1	15		26	$\overline{\text{S0}}$ ($\overline{\text{DEN}}$)		
AD0	16		25	QS0 (ALE)		
NMI	17		24	QS1 ($\overline{\text{INTA}}$)		
INTR	18		23	$\overline{\text{TEST}}$		
CLK	19		22	READY		
GND	20		21	RESET		

**Direct
Memory
Access**

Hold

**Hold
acknowledge**

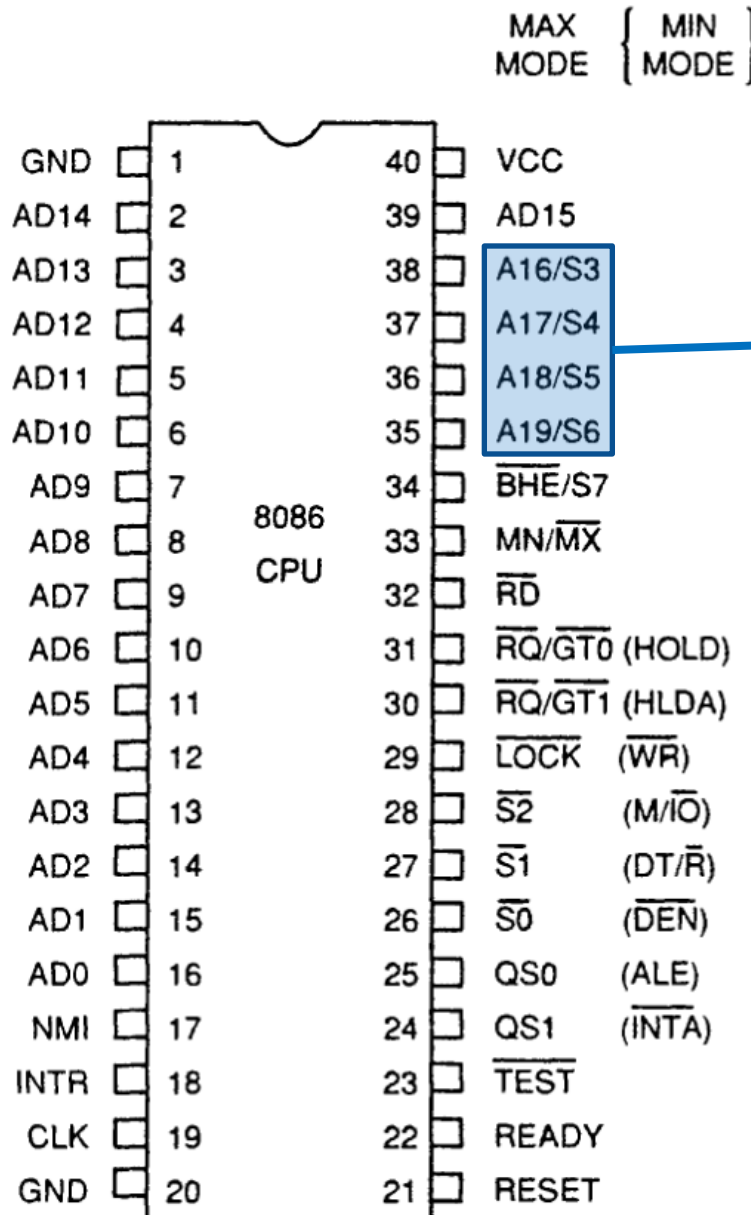
INTEL 8086 - Pin Details

S6: Logic 0.

S5: Indicates condition of IF flag bits.

S4-S3: Indicate which segment is accessed during current bus cycle:

S4	S3	Function
0	0	Extra segment
0	1	Stack segment
1	0	Code or no segment
1	1	Data segment



Address/Status Bus

Address bits $A_{19} - A_{16}$ & Status bits $S_6 - S_3$

INTEL 8086 - Pin Details

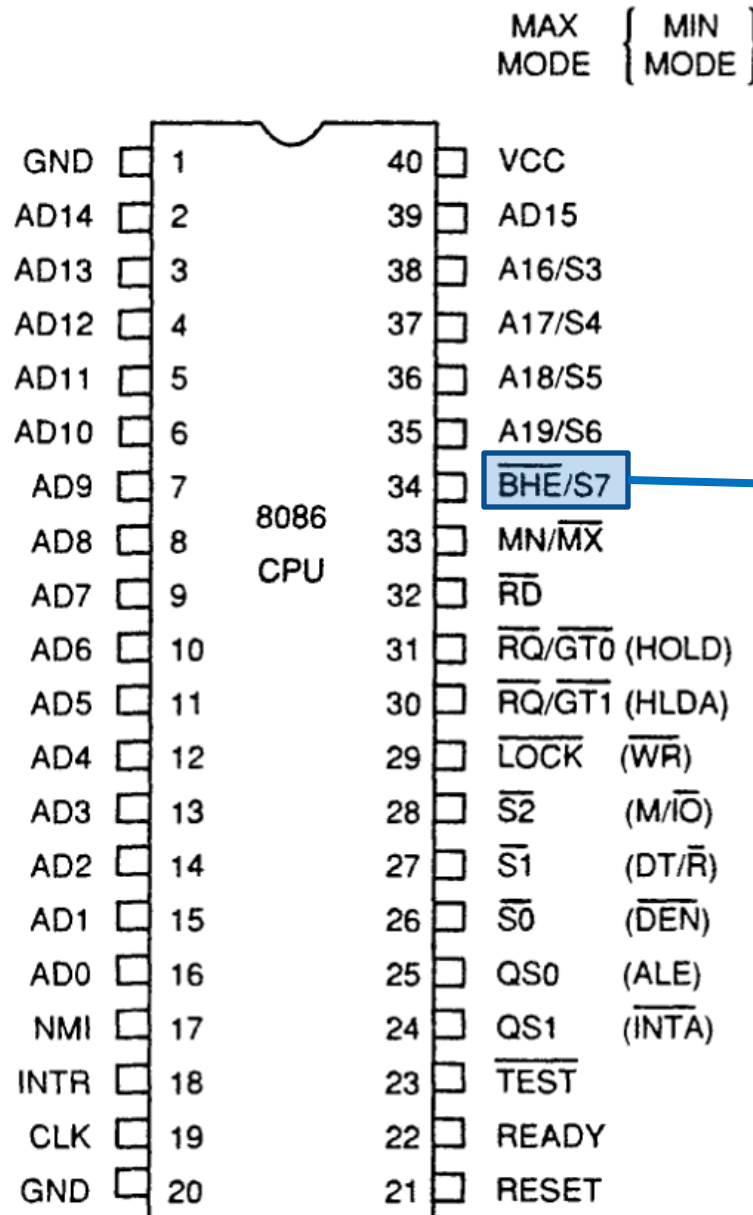
BHE#, A₀:

0,0: Whole word
(16-bits)

0,1: High byte
to/from odd address

1,0: Low byte
to/from even address

1,1: No selection

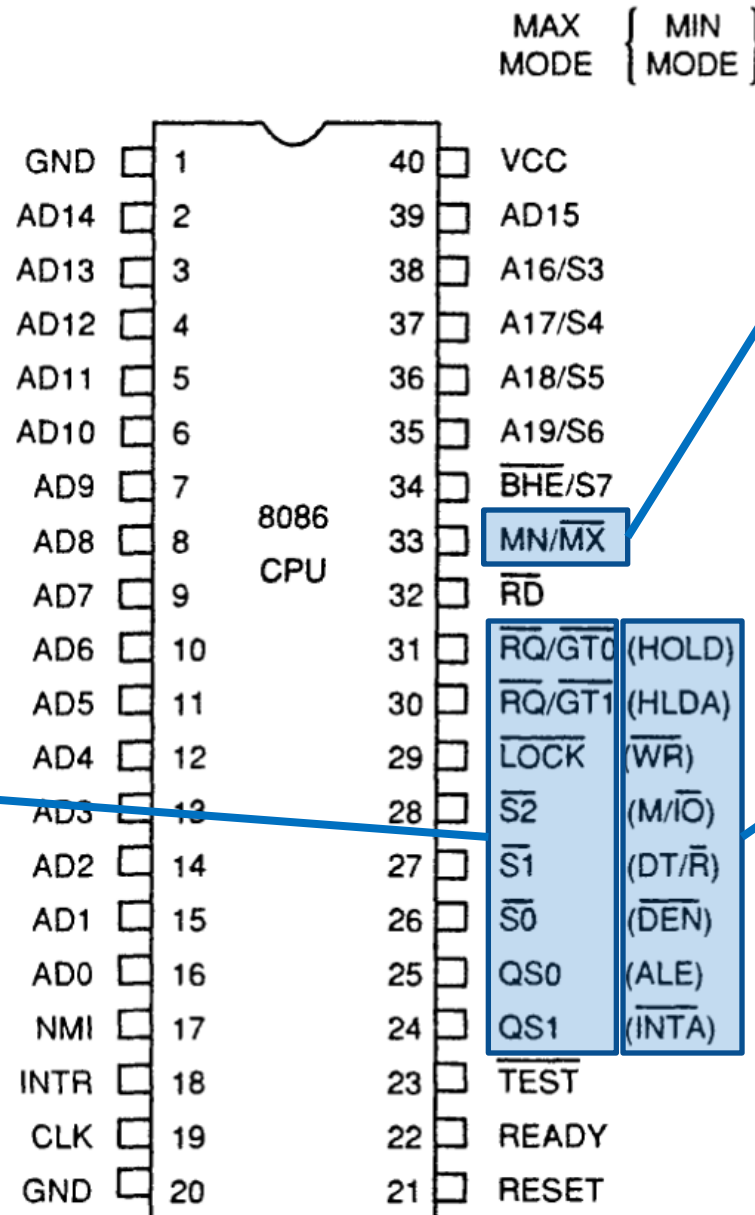


Bus High Enable/S7

Enables most significant data bits D₁₅ – D₈ during read or write operation.

S₇: Always 1.

INTEL 8086 - Pin Details



Min/Max mode

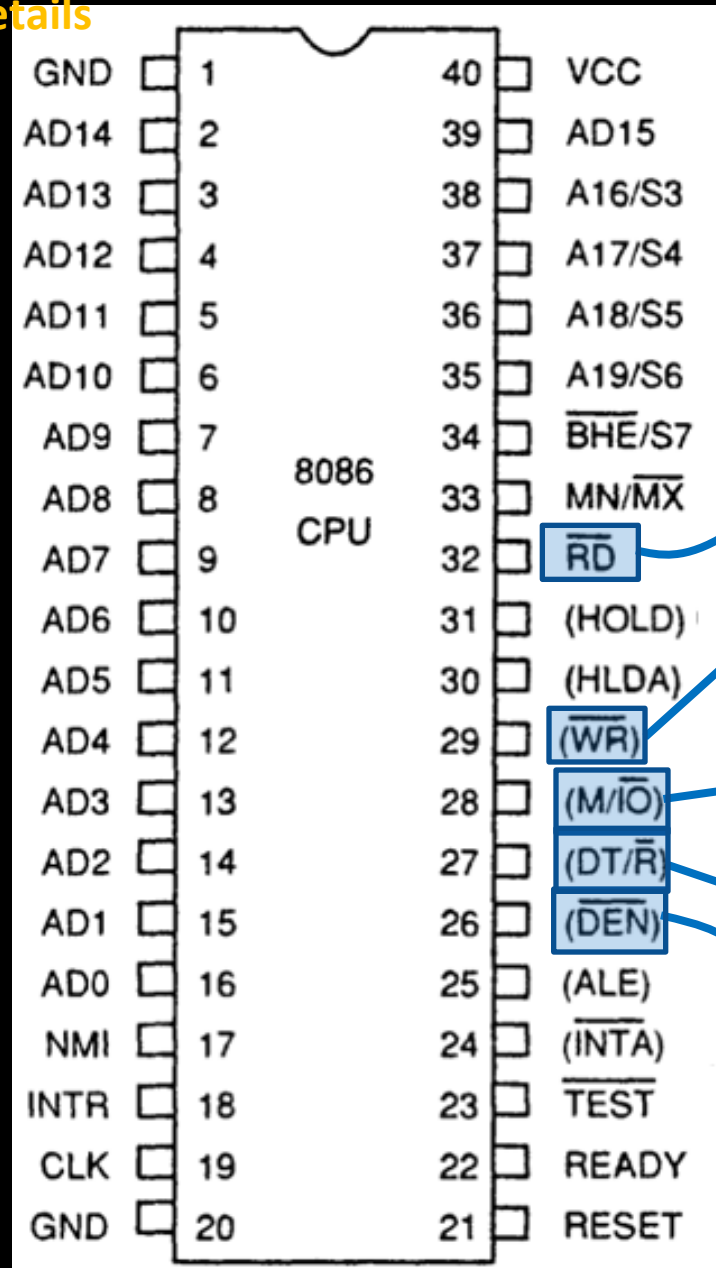
Minimum Mode: +5V

Maximum Mode: 0V

Maximum Mode Pins

Minimum Mode Pins

Minimum Mode- Pin Details



Read Signal

Write Signal

Memory or

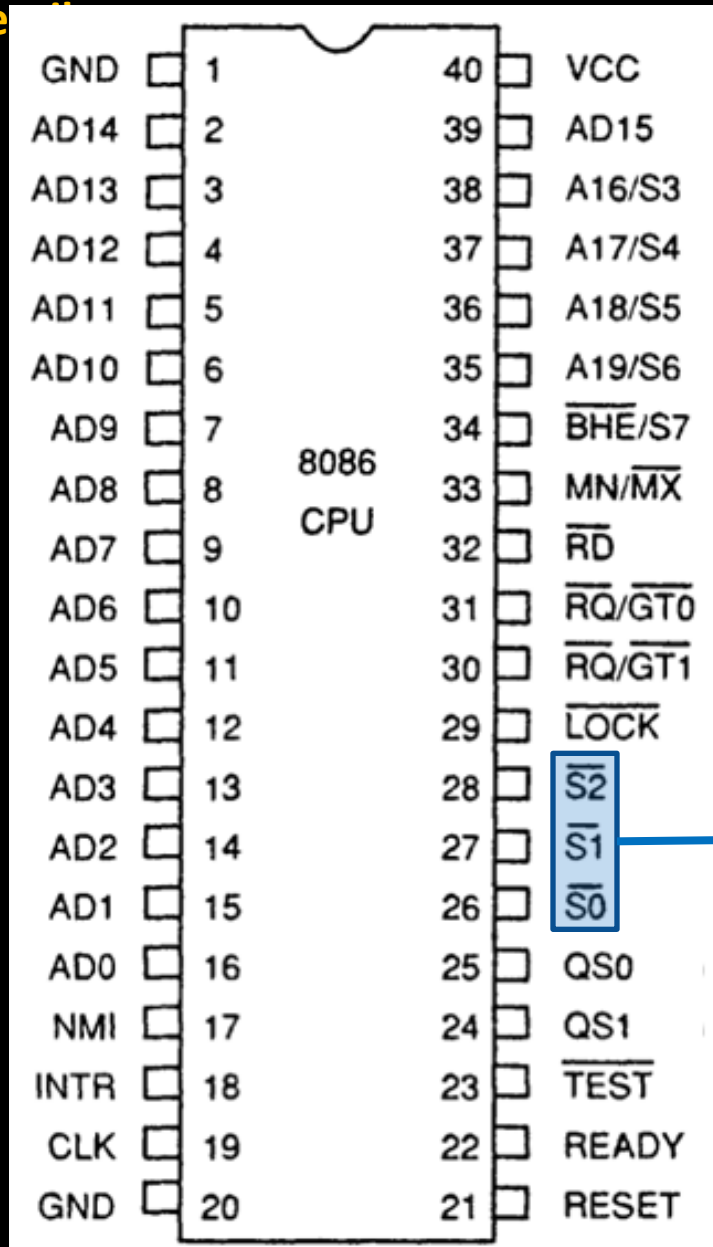
Data Transmit/Recei

Data Bus

Maximum Mode - Pin De

S2 S1 S0

000: INTA
 001: read I/O port
 010: write I/O port
 011: halt
 100: code access
 101: read memory
 110: write memory
 111: none -passive



Status Signal

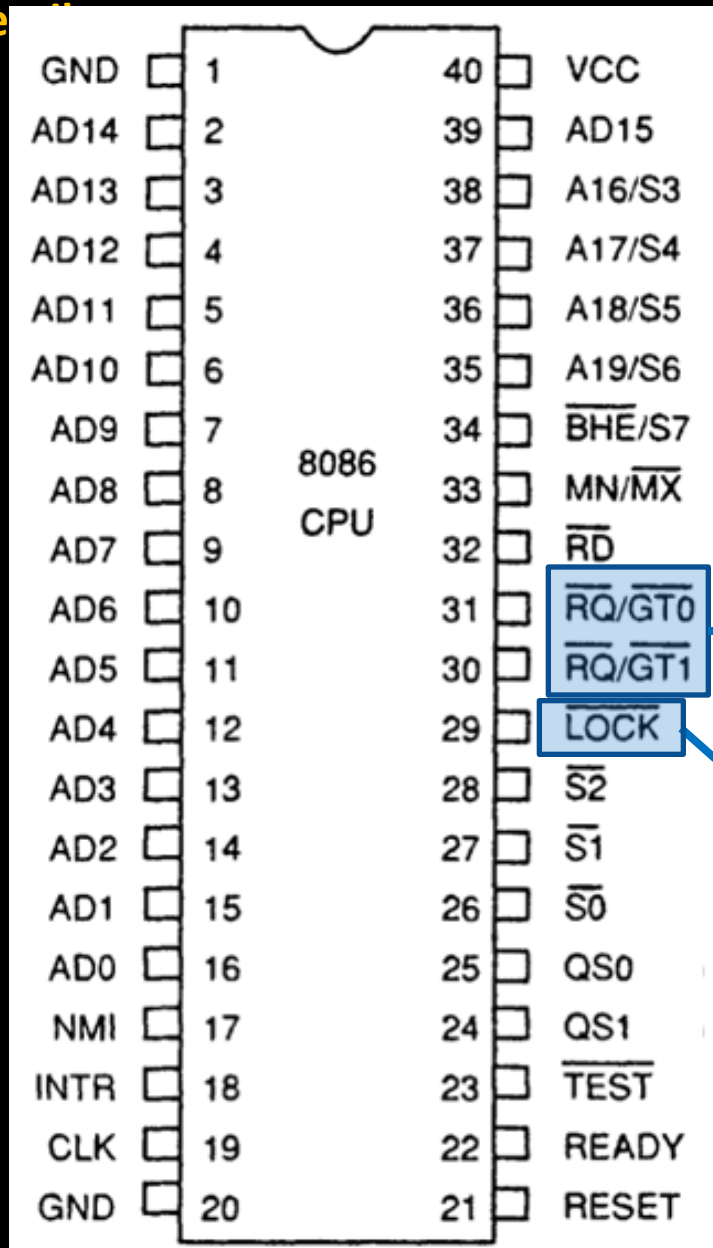
Inputs to 8288 to generate eliminated signals due to max mode.

Maximum Mode - Pin De

Lock Output

Used to lock peripherals off the system

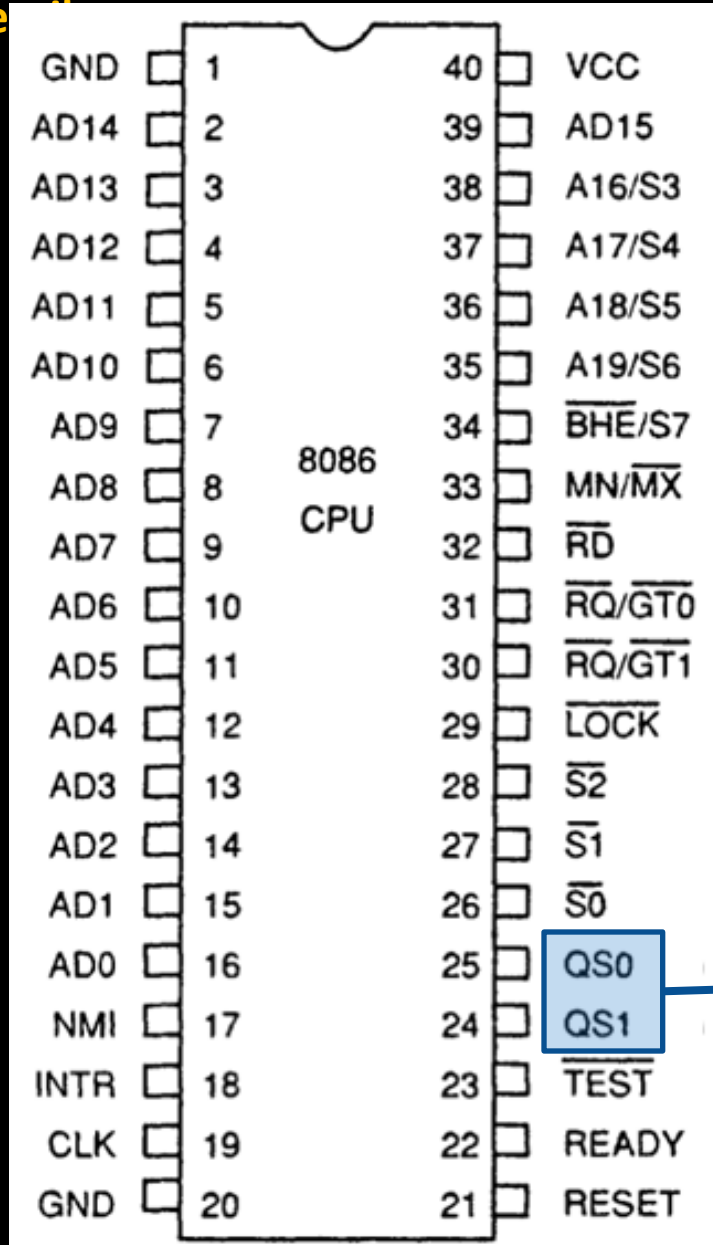
Activated by using the LOCK: prefix on any instruction



DMA
Request/Grant

Lock Output

Maximum Mode - Pin De



QS1 QS0

00: Queue is idle

01: First byte of opcode

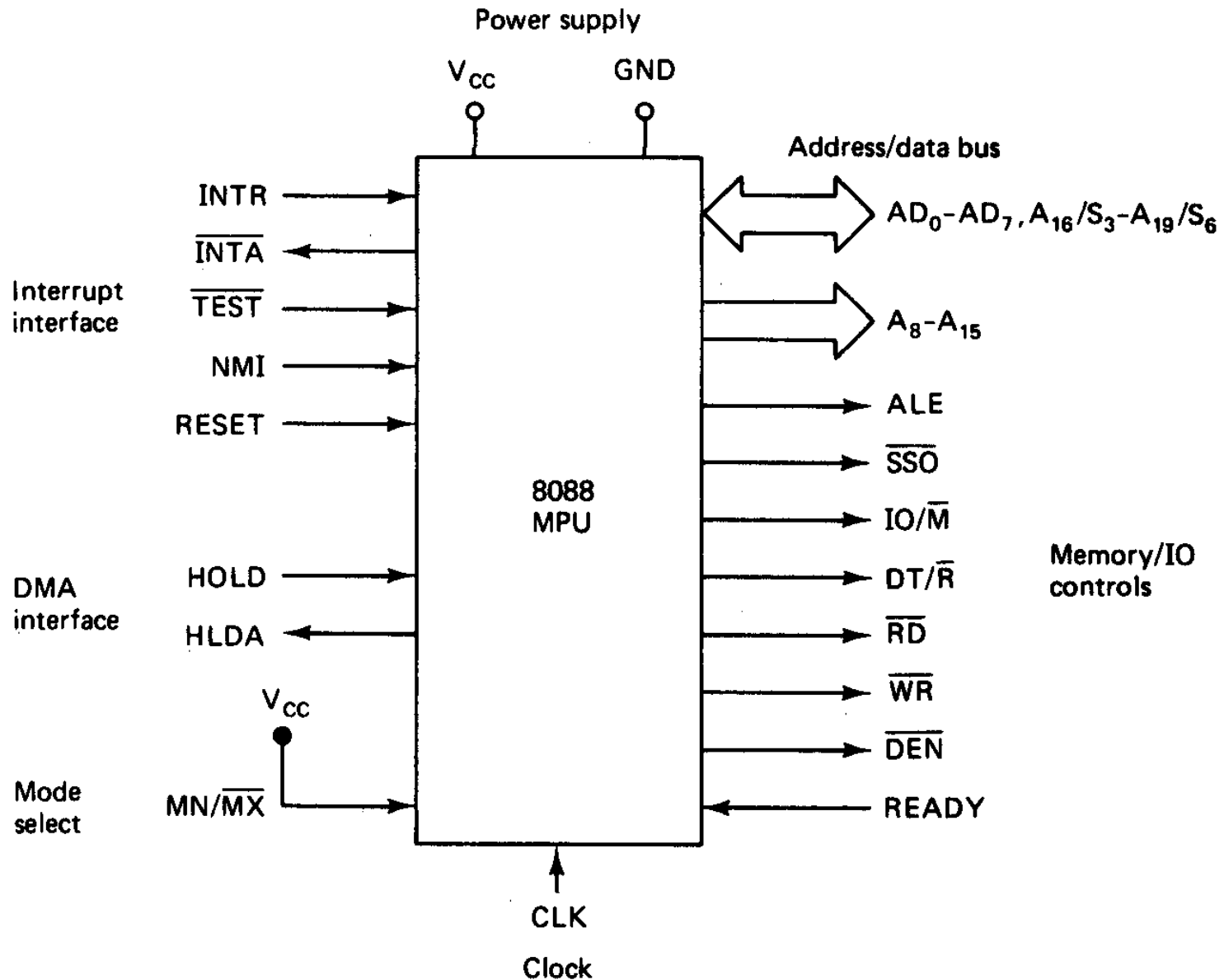
10: Queue is empty

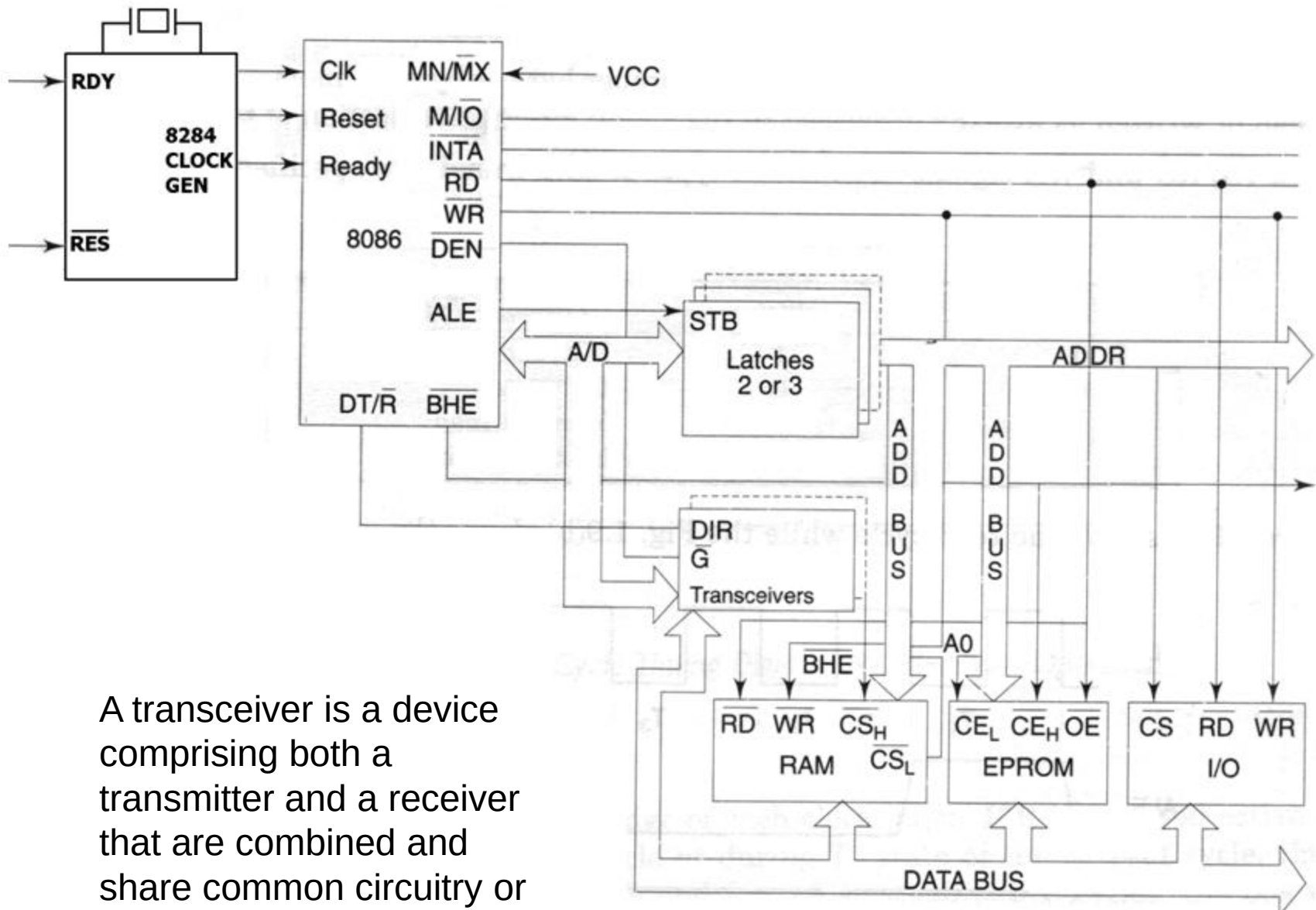
11: Subsequent byte of opcode

Queue Status

Used by numeric coprocessor (8087)

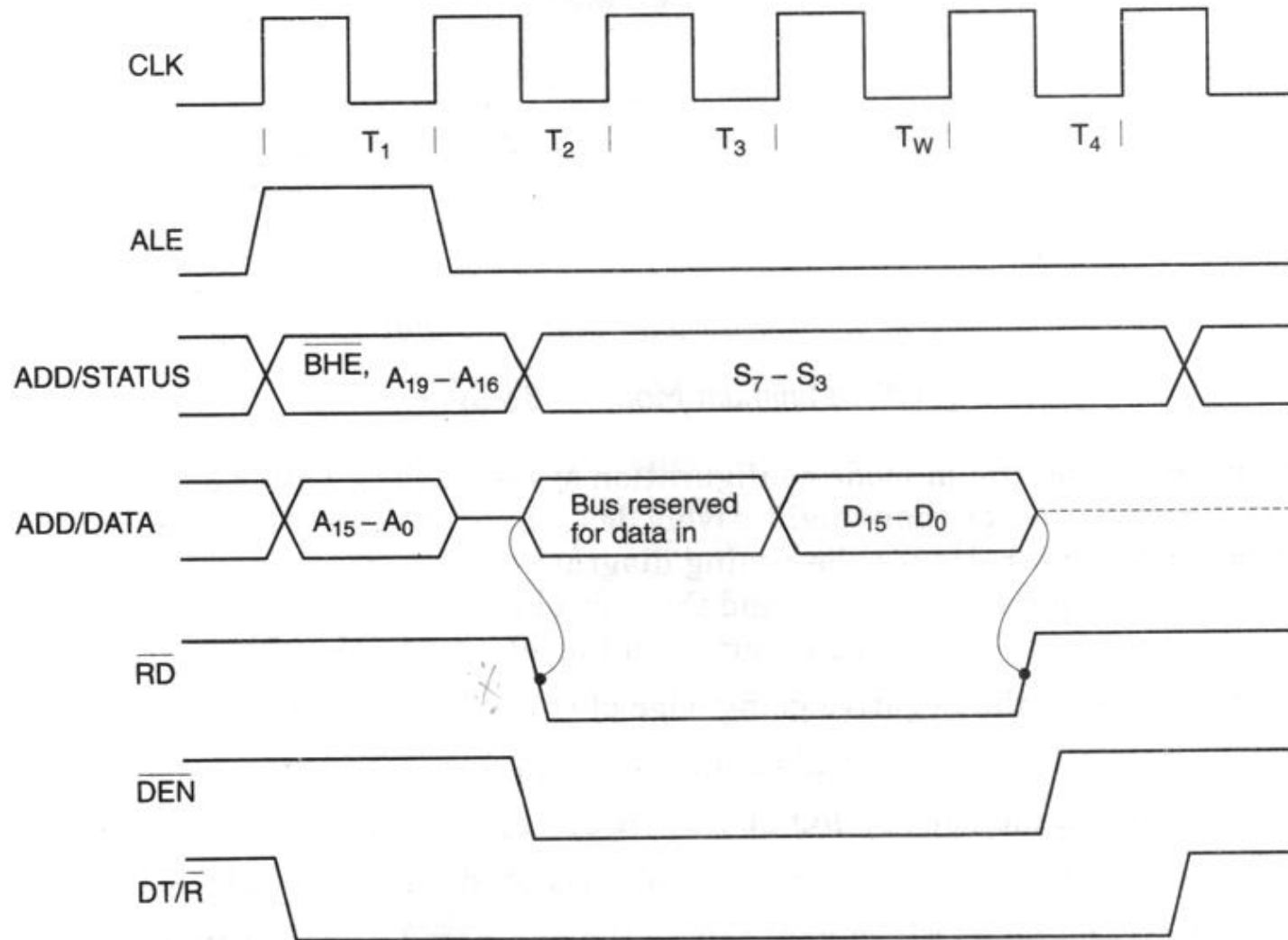
Minimum Mode 8086 System





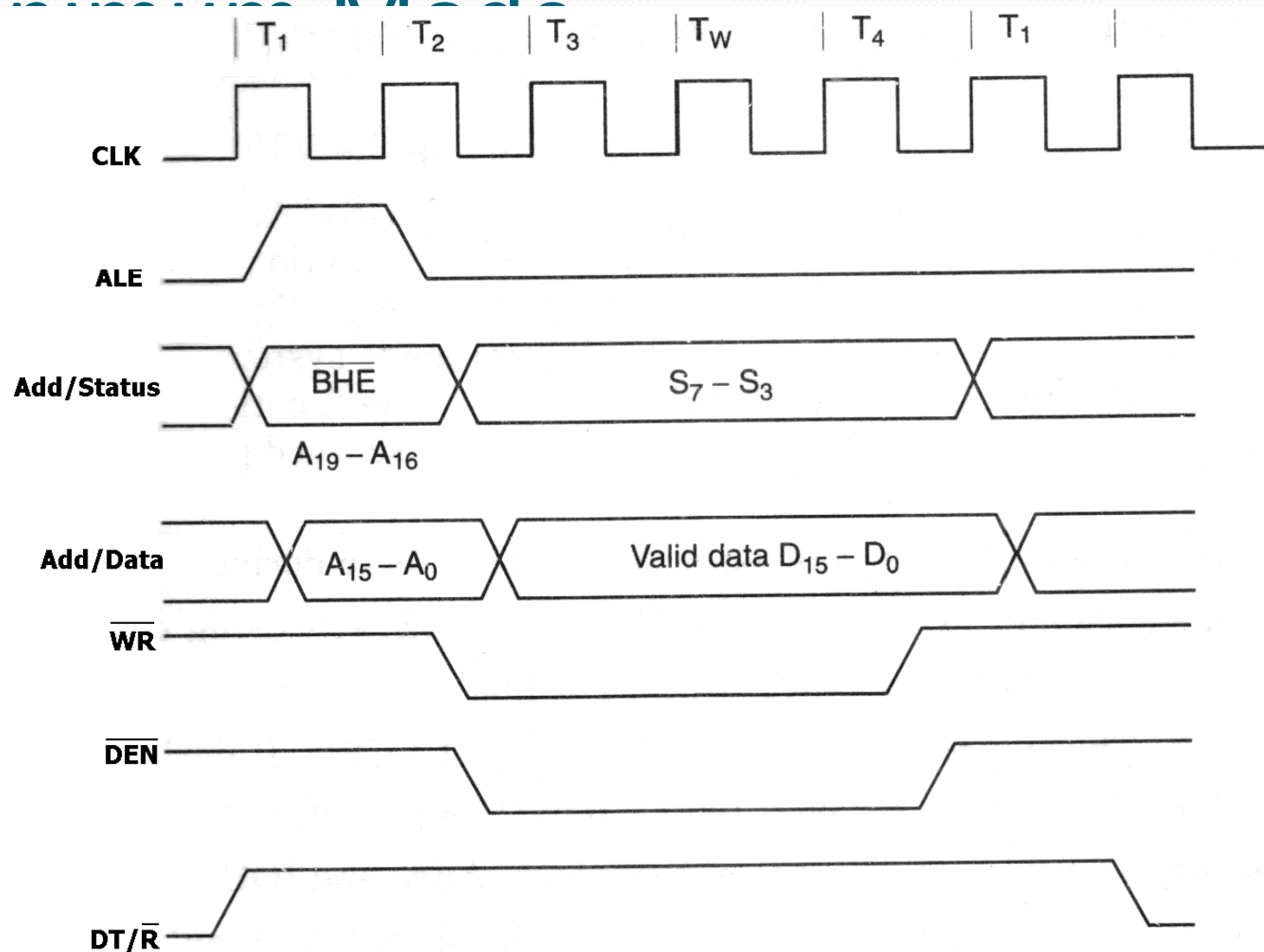
A transceiver is a device comprising both a transmitter and a receiver that are combined and share common circuitry or a single housing

'Read' Cycle timing Diagram for Minimum Mode



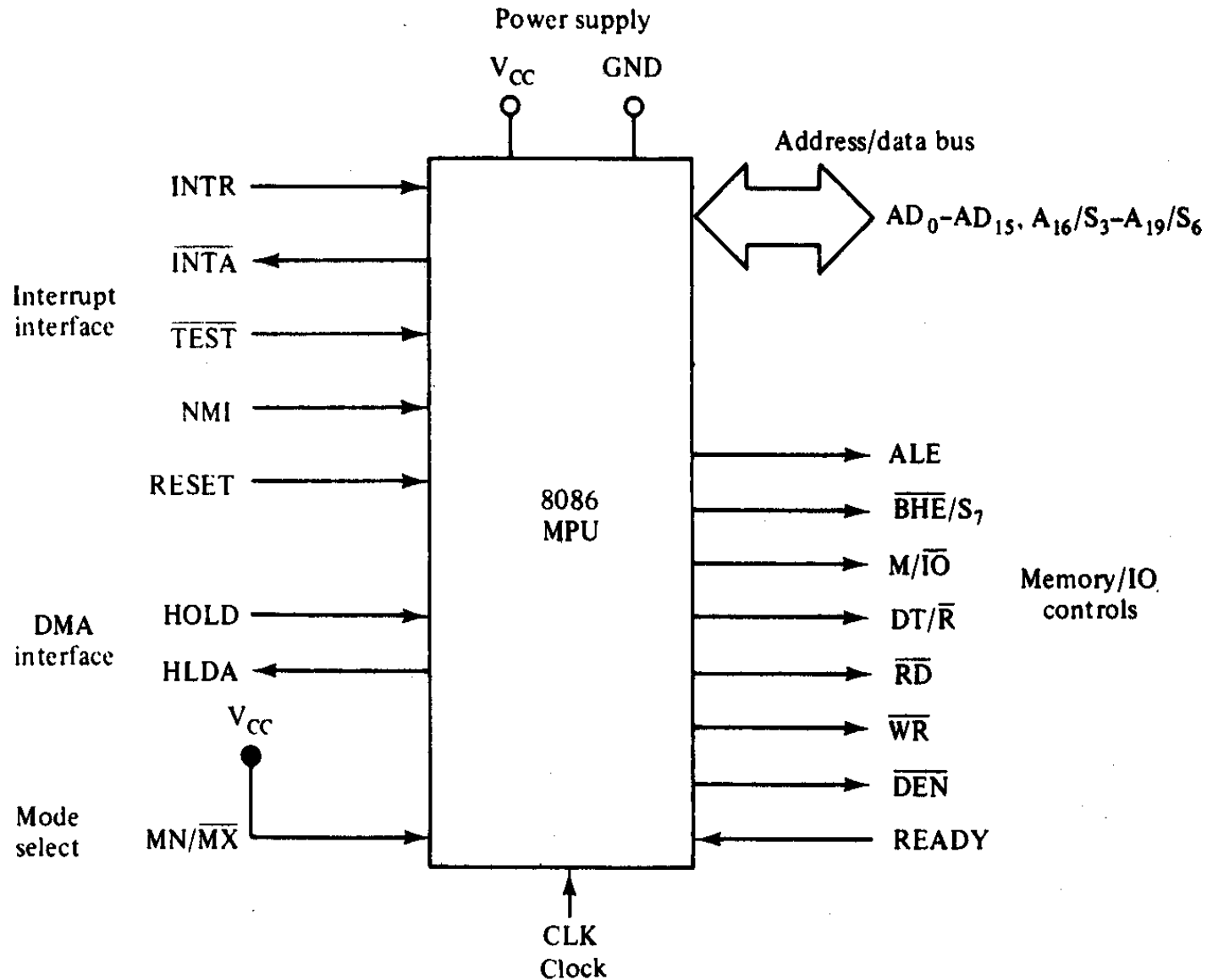
'Write' Cycle timing Diagram for

Minimally Mode

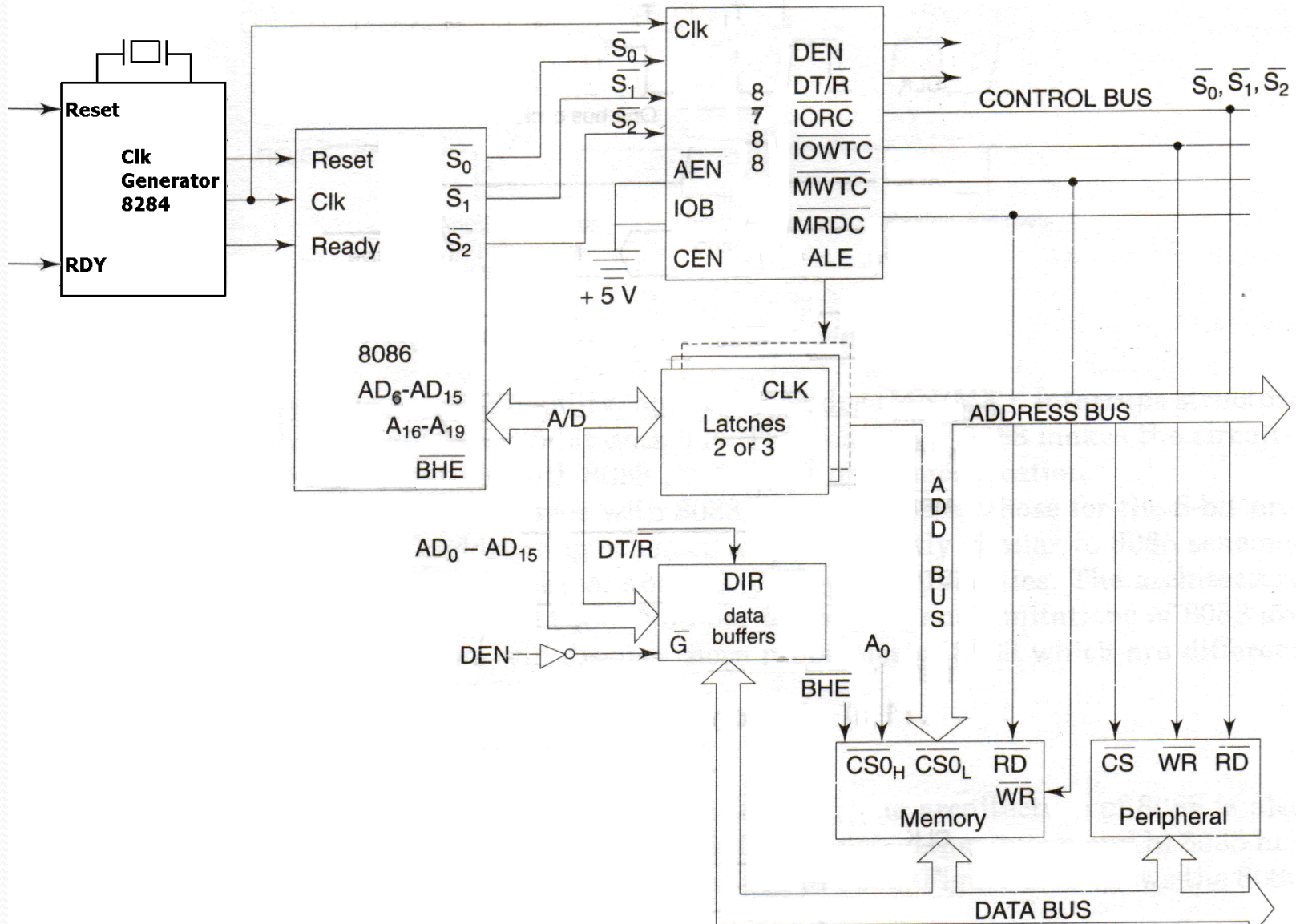


	MIN MODE	MAX MODE
1	It is a uniprocessor mode . 8086 is the only processor in the circuit.	It is a multiprocessor mode . Along with 8086, there can be other processors like 8087 and 8089 in the circuit.
2	Here MN/ $\overline{\text{MX}}$ is connected to Vcc .	Here MN/ $\overline{\text{MX}}$ is connected to Ground .
3	ALE for the latch is given by 8086 itself.	As there are multiple processors, ALE for the latch is given by 8288 bus controller .
4	$\overline{\text{DEN}}$ and DT/ $\overline{\text{R}}$ for the transreceivers are given by 8086 itself.	As there are multiple processors, DEN and DT/ $\overline{\text{R}}$ for the transreceivers is given by 8288 bus controller .
5	Direct control signals like M/ $\overline{\text{IO}}$, $\overline{\text{RD}}$ and $\overline{\text{WR}}$ are produced by 8086 itself.	Instead of control signals, all processors produce status signals $\overline{\text{S}}_2$, $\overline{\text{S}}_1$ and $\overline{\text{S}}_0$
6	Control signals M/ $\overline{\text{IO}}$, $\overline{\text{RD}}$ and $\overline{\text{WR}}$ are decoded by a 3:8 decoder IC 74138.	Status signals $\overline{\text{S}}_2$, $\overline{\text{S}}_1$ and $\overline{\text{S}}_0$ require special decoding are decoded by 8288 bus controller .
7	$\overline{\text{INTA}}$ for interrupt acknowledgement is produced by 8086.	$\overline{\text{INTA}}$ for interrupt acknowledgement is produced by 8288 Bus Controller .
8	Bus request are grant is handled using HOLD and HLDA signals.	Bus request are grant is handled using RQ / $\overline{\text{GT}}$ signals.
9	Since 74138 does not independently generate any signals, it does not need a CLK .	Since 8288 independently generates control signals, it needs a CLK from 8284 clock generator.
10	The circuit is simpler but does not support multiprocessing .	The circuit is more complex but supports multiprocessing .

Maximum Mode 8086 System

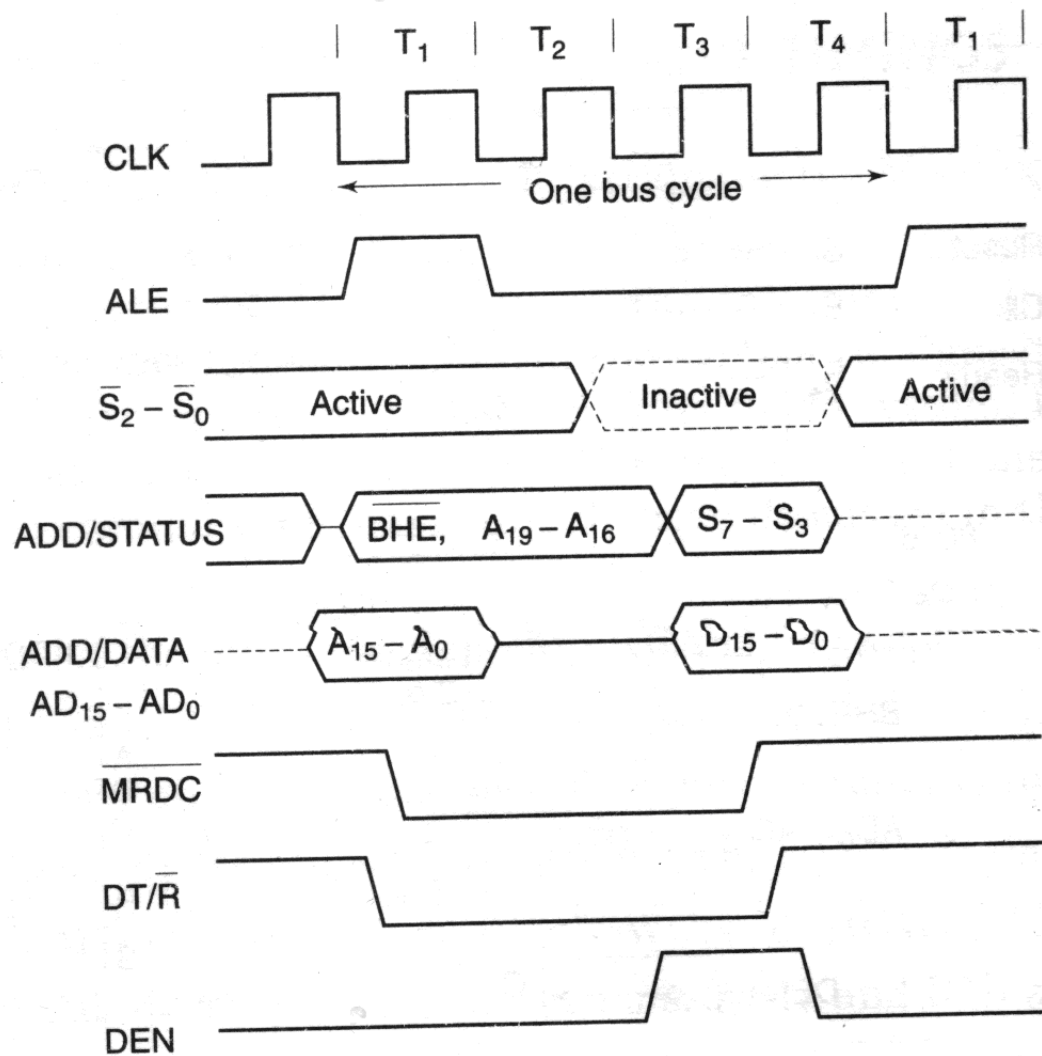


Maximum Mode 8086 System



- 
- # Maximum Mode 8086 System
- Here, either a numeric coprocessor of the type 8087 or another processor is interfaced with 8086.
 - The Memory, Address Bus, Data Buses are shared resources between the two processors.
 - The control signals for Maximum mode of operation are generated by the Bus Controller chip 8788.
 - The three status outputs So^* , $S1^*$, $S2^*$ from the processor are input to 8788.
 - The outputs of the bus controller are the Control Signals, namely DEN, DT/R^* , $IORC^*$, $IOWTC^*$, $MWTC^*$, $MRDC^*$, ALE etc.

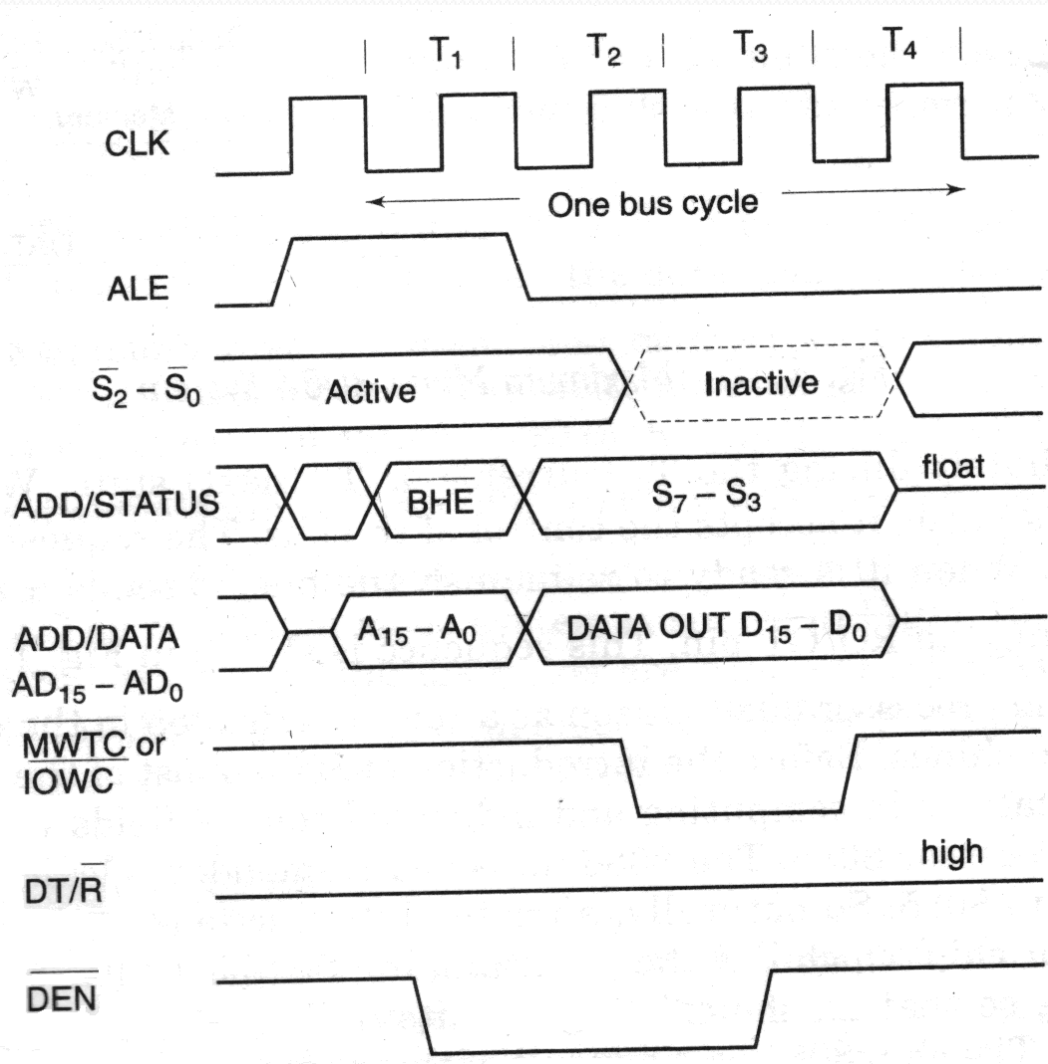
Memory Read timing in Maximum Mode



$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	Function
0	0	0	Interrupt acknowledge
0	0	1	I/O read
0	1	0	I/O write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
1	1	0	Memory write
1	1	1	Passive

TABLE 8-6 Bus control functions generated by the bus controller (8288) using $\overline{S_2}$, $\overline{S_1}$, and $\overline{S_0}$

Memory Write timing in Maximum Mode



$\overline{S}_2$	$\overline{S}_1$	$\overline{S}_0$	Function
0	0	0	Interrupt acknowledge
0	0	1	I/O read
0	1	0	I/O write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
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1	1	1	Passive

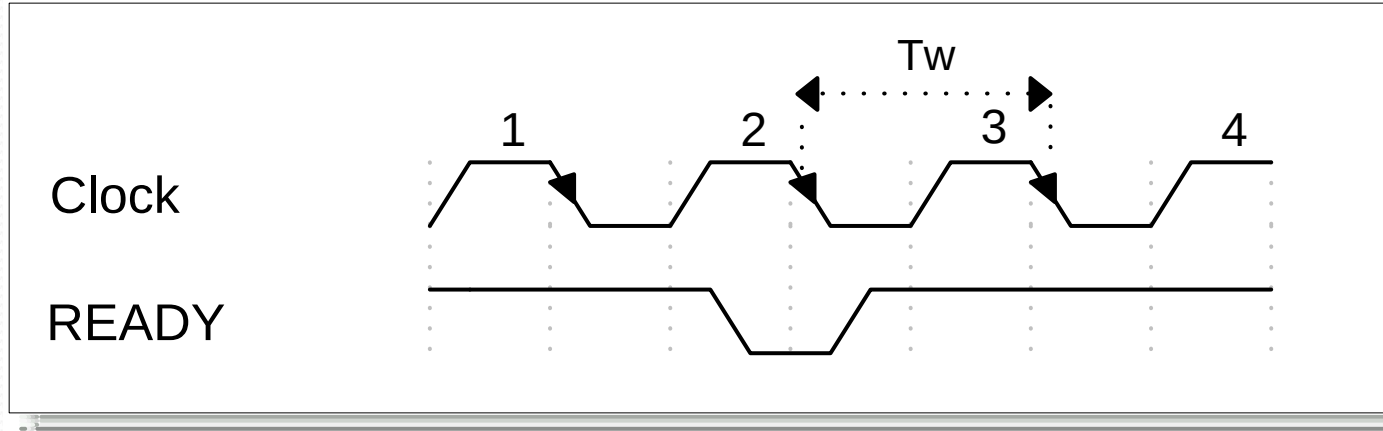
TABLE 8-6 Bus control functions generated by the bus controller (8288) using $\overline{S}_2$, $\overline{S}_1$, and $\overline{S}_0$

8086 Control Signals

1. ALE
2. BHE
3. M/IO
4. DT/R
5. RD
6. WR
7. DEN

- 
- ## • TEST pin of 8086
- Used in conjunction with the WAIT instruction in multiprocessing environments.
 - This is input from the 8087 coprocessor.
 - During execution of a wait instruction, the CPU checks this signal.
 - If it is low, execution of the signal will continue; if not, it will stop executing.

WAIT State

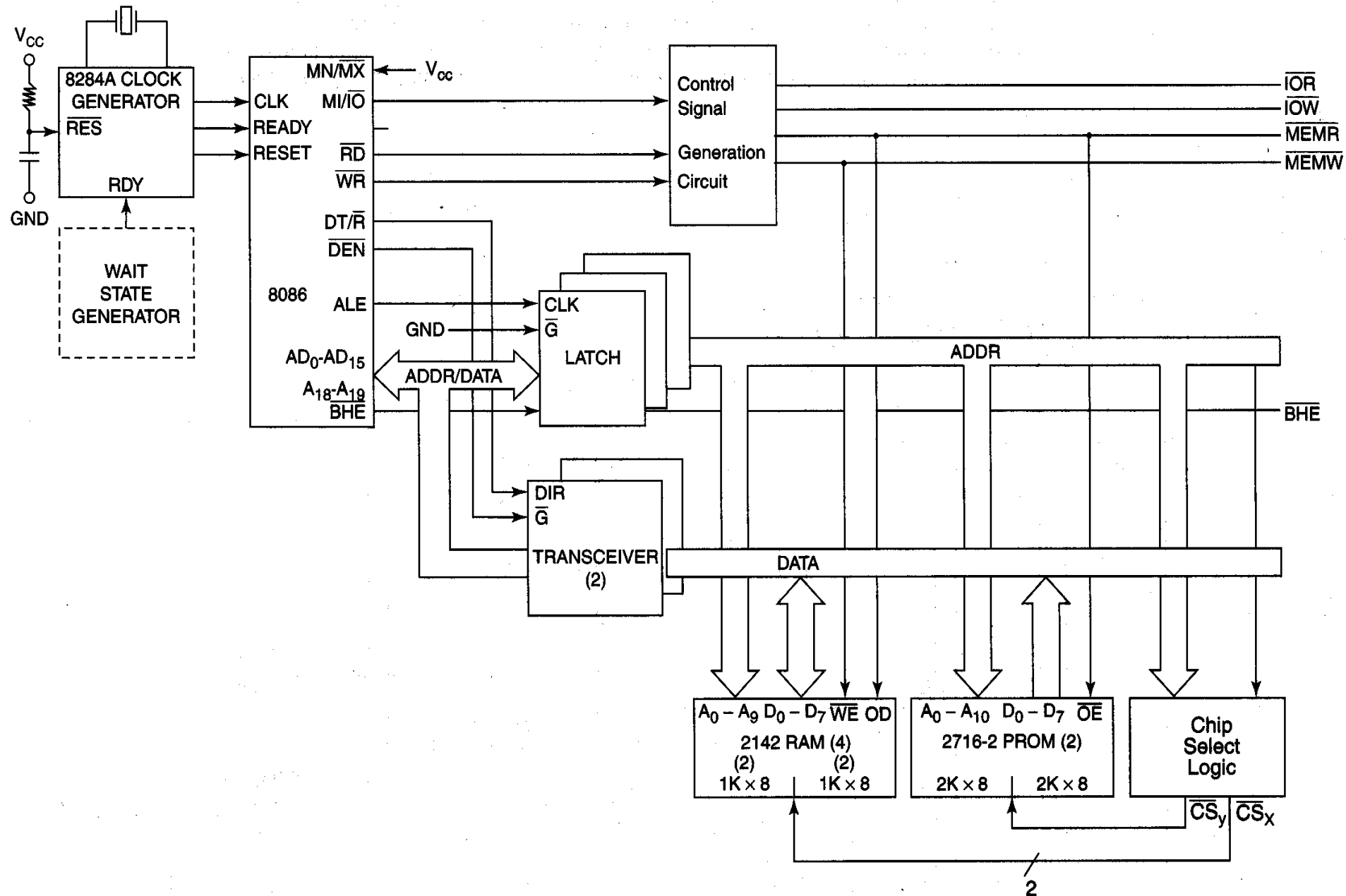


- A **wait state (T_w)** is an extra clocking period, inserted between **T_2** and **T_3** , to lengthen the bus cycle, allowing slower memory and I/O components to respond.
- The **READY** input is sampled at the end of **T_2** , and again, if necessary in the middle of T_w . **If READY is 'o' then a T_w is inserted.**

8086 System Memory Circuitry

1. Minimum Mode System Memory Circuitry
2. Maximum Mode System Memory Circuitry

Minimum Mode System Memory Circuitry



Maximizing Mode System Memory Circuitry

